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Learn the Basics – Python

A Professional Introduction to Python Programming

This clean, well-structured guide covers the essential fundamentals of Python. It is designed for beginners and serves as an excellent quick reference.

1. Hello, World!

This is the traditional first program in Python. The language is known for its simple, readable syntax and minimal boilerplate.

2. Variables and Types

Python is dynamically typed — you do not need to declare variable types. You can store numbers, strings, booleans, and more in variables.

3. Lists

Lists are ordered, mutable collections that can store different data types. They support adding, removing, and accessing elements by index.

4. Basic Operators

Basic operators include arithmetic operations such as addition, subtraction, multiplication, division, modulus, and exponentiation. They can also be used with strings and lists.

5. String Formatting

String formatting allows you to insert variables into text. Common methods include the older % operator and modern f-strings.

6. Basic String Operations

Common string operations include finding length, converting to uppercase/lowercase, slicing, checking substrings, and replacing text.

7. Conditions

Conditions (if-elif-else statements) allow the program to make decisions based on whether certain conditions are true or false. Logical operators such as and, or, and not are commonly used.

8. Loops

Loops are used to repeat blocks of code. The two main types are:

- **For loops** — used to iterate over a sequence (like a list or range).
 - **While loops** — repeat as long as a condition remains true.
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Practice Recommendations:

- Run examples in any Python environment.
 - Modify values and experiment with the concepts.
 - Consistent daily practice is the best way to master Python.
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Document Summary

This professional Markdown document demonstrates excellent note-taking structure and formatting. It is clean, consistent, and ready for printing or PDF conversion.

Best Practices Applied:

- Clear numbered sections
- Concise explanations
- Professional layout and spacing

Created for effective learning and documentation practice.

This document can be easily exported to PDF using Typora, Obsidian, or any Markdown-to-PDF tool.